



ITE231 | WEEK 5 ACTIVITY

API Integration & HTTP Requests

EverCare OpenStreetMap Hospital Finder

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Prepared: August 15, 2026

Due: August 17, 2026 at 7:00 PM

IMPLEMENTATION OUTCOME

EverCare already contains a complete external API integration. Live verification returned HTTP 200 from both GET and POST requests, with 15 Nominatim search records and 75 Overpass hospital elements. The app displayed the processed results on the map and in the appointment workflow.

GitHub repository (submitted instead of a source-code ZIP): <https://github.com/Brenttttttt/evercare>

This document provides the API purpose, endpoints, request methods, JSON-processing evidence, live UI screenshots, and verification results required for Week 5.

1. Integration Overview

The OpenStreetMap Hospital Finder gives caregivers a practical way to find nearby hospitals, search by hospital or place name, see returned locations on a map, and select a hospital while creating an appointment. The same hospital finder is also available from EverCare's Emergency page.

How the integration works

1. The user searches by name or requests nearby hospitals from the EverCare interface.
2. HospitalFinderService sends a rate-limited HTTP GET or a form-encoded HTTP POST request.
3. The service validates and decodes the JSON response, maps records to HospitalLocation objects, removes duplicates, and calculates distance when a device position is available.
4. HospitalFinderScreen displays markers, names, addresses, distances, loading states, and retry guidance. Selecting a result fills the appointment hospital and address fields.

APIs and endpoints

API	Endpoint	Method	Purpose
Overpass API	https://overpass-api.de/api/interpreter	POST	Find hospitals within a 15 km radius; two fallback providers are configured.
Nominatim Search	https://nominatim.openstreetmap.org/search	GET	Search Philippine hospitals and places after the user submits a query.
Nominatim Reverse	https://nominatim.openstreetmap.org/reverse	GET	Resolve a missing address for a selected result.
Photon	https://photon.komoot.io/api	GET	Provide debounced search-as-you-type hospital suggestions.
OSM Tiles	https://tile.openstreetmap.org/{z}/{x}/{y}.png	GET	Render the in-app map with visible OpenStreetMap attribution.

WHY THIS API IS USEFUL

The integration reduces manual typing, makes hospital locations easier to review, and connects appointment planning with nearby-care discovery. It does not require a paid API key.

2. Week 5 Requirements Compliance

Requirement	EverCare evidence
Integrate at least one API	OpenStreetMap hospital discovery uses Overpass, Nominatim, Photon, and OSM tiles.
HTTP GET requests	Nominatim search/reverse and Photon autocomplete use <code>http.Client.get</code> with headers and timeouts.
HTTP POST requests	Overpass uses <code>http.Client.post</code> with a form-encoded data field containing the bounded hospital query.
Parse JSON responses	<code>jsonDecode</code> validates object, list, and GeoJSON response shapes before typed mapping.
Display dynamic API data	Hospital markers, result counts, names, addresses, and distances are rendered in the finder.
Handle loading and invalid responses	The UI provides progress, empty, permission, API error, and Try Again states. The service reports malformed JSON honestly.
Brief API explanation	Purpose, endpoints, methods, processing, limitations, and tests are documented here.
Updated source code	The GitHub repository link is supplied in place of a source ZIP, as requested.

Request safeguards

- Identifiable EverCare User-Agent and JSON Accept headers.
- One-second serialization for public geocoder requests and 900 ms UI debounce for typing.
- Bounded result limits, a 15 km nearby radius, and in-memory caching.
- Timeouts plus fallback Overpass providers for HTTP 429, server failures, and network errors.
- Philippines-only filtering for submitted searches and autocomplete suggestions.

DATA AND SAFETY NOTE

The submitted evidence uses a generic public Guiguinto location and contains no patient data, credentials, keys, or precise user location. OpenStreetMap results may be incomplete or outdated and do not confirm hospital availability, capacity, contact accuracy, or quality of care.

3. HTTP POST Implementation

EverCare builds an Overpass QL query for amenity=hospital or healthcare=hospital, sends it as application/x-www-form-urlencoded, verifies the HTTP status, then decodes the JSON object and reads its elements array.

```

HTTP POST - Overpass nearby-hospital request
lib/services/hospital_finder_service.dart

294   final query =
295   '...',
296   [out:json][timeout:25];
297   (
298     nwr["amenity"="hospital"](around:$radius,$latitude,$longitude);
299     nwr["healthcare"="hospital"](around:$radius,$latitude,$longitude);
300   );
301   out center tags;
302   '...';
303
304   final response = await _requestOverpassWithFallback(query);
305
306   try {
307     final body = jsonDecode(utf8.decode(response.bodyBytes));
308     if (body is! Map<String, dynamic>) {
309       throw const FormatException('Expected an object response.');
```

Figure 1. Actual EverCare source showing the Overpass POST request, JSON decoding, and successful-status check.

Live result: HTTP 200 with 75 OpenStreetMap elements around the public Guiguinto demonstration center. EverCare deduplicates, sorts, and displays up to 25 nearby results.

4. HTTP GET and JSON Processing

A submitted search builds a Nominatim URL with jsonv2 output, Philippine country filtering, address details, a bounded result limit, and deduplication. The response must be a JSON array before each item is mapped to a HospitalLocation.

```

HTTP GET - Nominatim search and JSON parsing
lib/services/hospital_finder_service.dart

378 Future<List<HospitalLocation>> _loadSearch(String query) async {
379   final uri = _nominatimEndpoint.replace(
380     queryParameters: {
381       ..._nominatimEndpoint.queryParameters,
382       'format': 'jsonv2',
383       'q': _hospitalSearchQuery(query),
384       'countrycodes': 'ph',
385       'addressdetails': '1',
386       'limit': '15',
387       'dedupe': '1',
388     },
389   );
390   final response = await _rateLimitedNominatimRequest(uri);
391   _ensureSuccessful(response, serviceName: 'OpenStreetMap search service');
392
393   try {
394     final body = jsonDecode(utf8.decode(response.bodyBytes));
395     if (body is! List) {
396       throw const FormatException('Expected a list response.');
```

Figure 2. Actual EverCare source showing Nominatim GET parameters, status validation, jsonDecode, and typed mapping.

Live result: HTTP 200 with 15 hospital and place records for the submitted query “hospital, Bulacan, Philippines.”

5. Retrieved API Data in EverCare

The nearby flow uses device location only after permission is available, performs the Overpass POST, and displays processed results with map markers and distance labels.

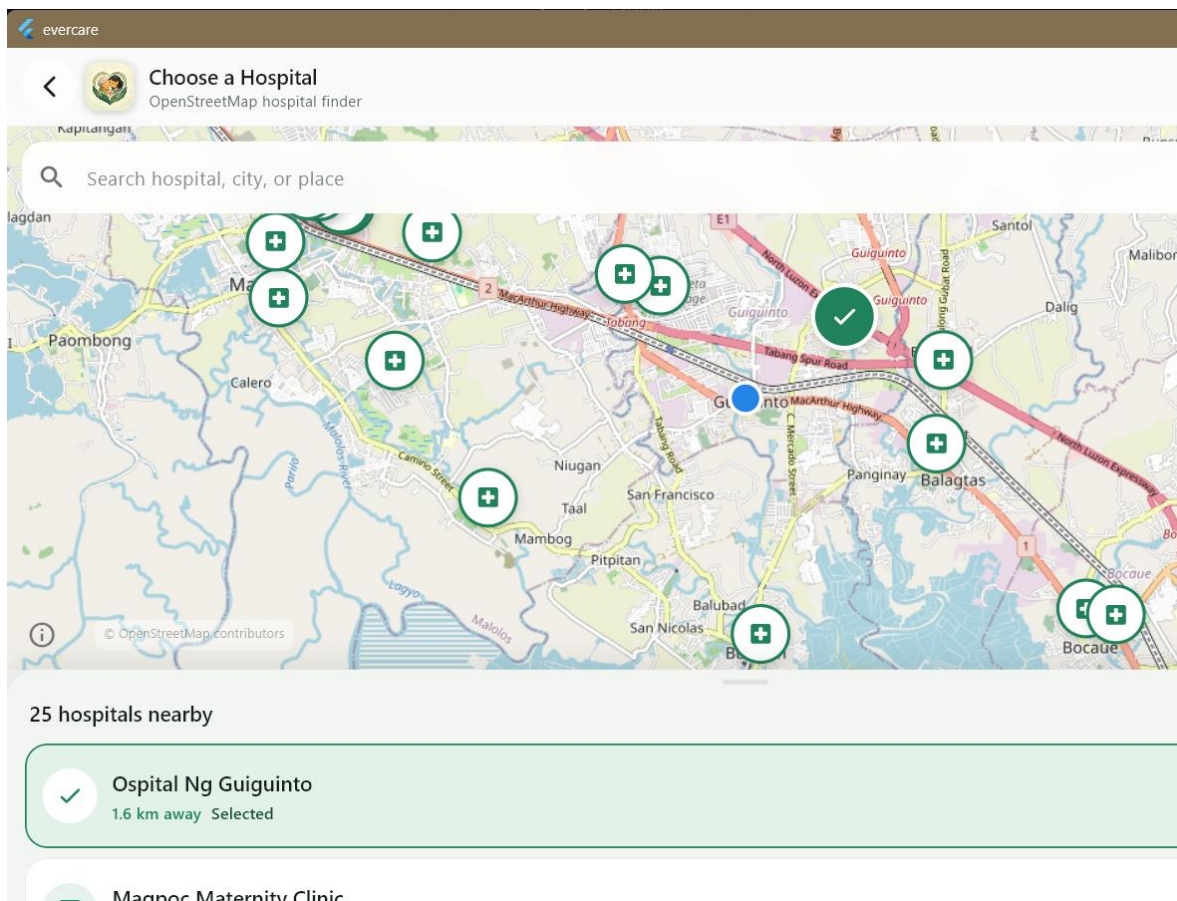


Figure 3. Live EverCare hospital finder showing 25 nearby processed results, map markers, OpenStreetMap attribution, and a selected hospital.

MEANINGFUL DISPLAY

Raw JSON is not shown directly to the caregiver. EverCare converts it into readable hospital cards, coordinates, addresses when available, distance labels, and interactive map markers.

6. Search and Appointment Workflow

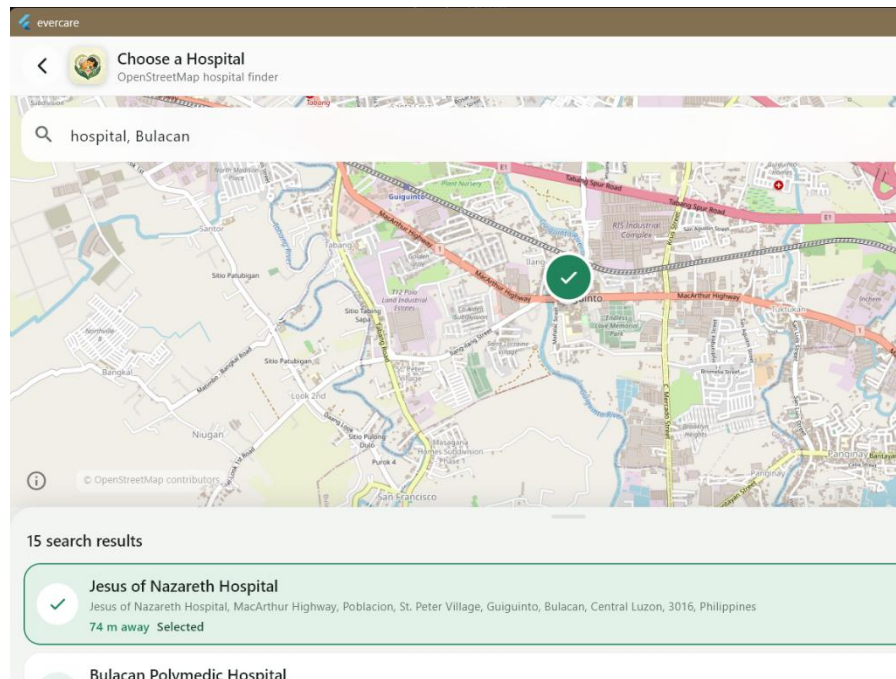


Figure 4. Submitted GET search displaying 15 named Bulacan results, including a selected result and distance.

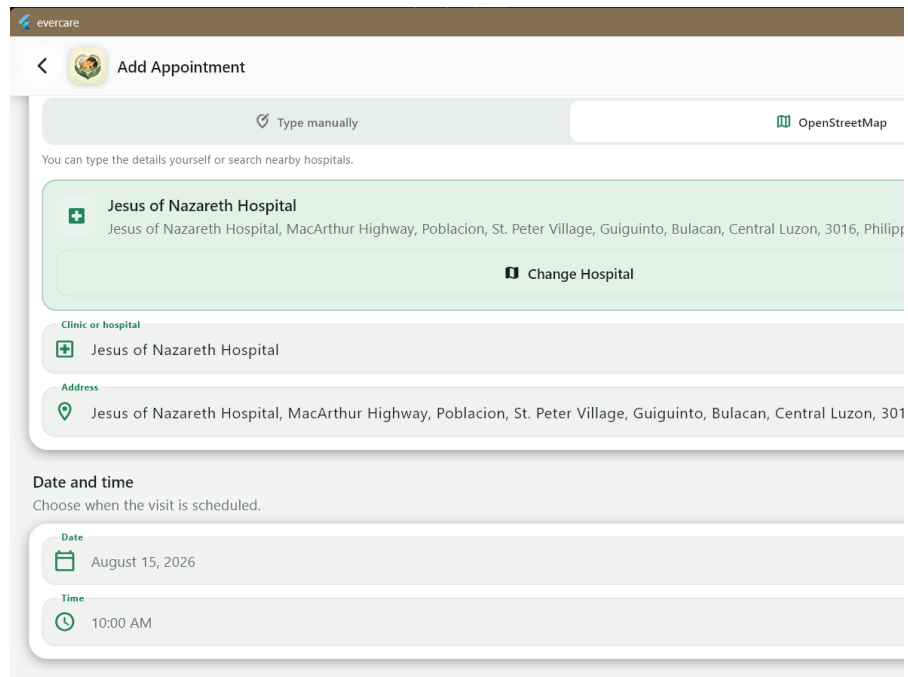


Figure 5. The selected API result automatically fills both the Clinic or hospital and Address fields.

7. JSON Response Sample

The following excerpt was generated from the live Overpass HTTP 200 response. The complete response and a separate Nominatim JSON array are included in `api_samples`.

```

Live Overpass JSON response - HTTP 200
docs\week 5\api_samples\03_overpass_nearby_excerpt.json

1  {
2    "evidence": "Live public OpenStreetMap data captured for EverCare Week 5",
3    "endpoint": "https://overpass-api.de/api/interpreter",
4    "method": "POST",
5    "capturedAt": "2026-08-15T10:15:26.7746832+08:00",
6    "publicDemoCenter": {
7      "latitude": 14.8333,
8      "longitude": 120.8833
9    },
10   "radiusMeters": 15000,
11   "resultCount": 75,
12   "elements": [
13     {
14       "type": "node",
15       "id": 589717017,
16       "lat": 14.7384585,
17       "lon": 120.9580669,
18       "tags": {
19         "amenity": "hospital",
20         "name": "B A Hospital",
21         "source": "GPS"
22       }
23     },
24     {
25       "type": "node",
26       "id": 597011693,
27       "lat": 14.906521,
28       "lon": 120.8695972,
29       "tags": {
30         "amenity": "hospital",
31         "healthcare": "hospital",
32         "name": "F M Cruz Ortopedic",
33         "source": "GPS"
34       }
35     },
36     {
37       "type": "node",
38       "id": 597116607,
39       "lat": 14.7445042,
40       "lon": 120.9110293,
41       "tags": {
42         "addr:street": "Tawiran Road",
43         "amenity": "hospital",
44         "name": "Tawiran Public Clinic"
45       }
46     }
47   ]
48 }

```

Figure 6. Genuine Overpass JSON excerpt containing the request method, result count, coordinates, hospital tags, and names.

Full GET sample: `api_samples/01_nominatim_search_response.json`

Full POST sample: `api_samples/02_overpass_nearby_response.json`

8. Loading, Error Handling, and Verification

Loading, empty, permission, and retry states

```
lib/screens/hospitals/hospital_finder_screen.dart

217 Widget _buildResults() {
218   if (_locationIssue != null && _hospitals.isEmpty) {
219     return _LocationHelp(
220       issue: _locationIssue!,
221       onRetry: _findNearby,
222       onSettings: _openRelevantSettings,
223     );
224   }
225   if (_error != null && _hospitals.isEmpty) {
226     return _FinderMessage(
227       icon: Icons.map_outlined,
228       title: 'Hospitals could not be loaded',
229       message: _error!,
230       actionLabel: 'Try Again',
231       onAction: _hasSearched ? _submitSearch : _findNearby,
232     );
233   }
234   if (_loading && _hospitals.isEmpty) {
235     return const Center(child: CircularProgressIndicator());
236   }
237   if (_hospitals.isEmpty) {
238     return _FinderMessage(
239       icon: Icons.local_hospital_outlined,
240       title: _hasSearched ? 'No hospitals found' : 'Find nearby hospitals',
241       message: _hasSearched
242         ? 'Try another hospital name, city, or province.'
243         : 'Allow location access to see hospitals near you, or search by name.',
244     );
245   }
}
```

Figure 7. Actual UI code for permission guidance, loading progress, empty results, API errors, and Try Again.

Verification results

- flutter analyze: PASS - no issues found.
- flutter test: PASS - all 99 tests passed.
- HospitalFinderService tests: PASS - all 10 tests passed.
- Windows debug build: PASS - evercare.exe built successfully.
- Live Nominatim GET: HTTP 200 - 15 JSON records.
- Live Overpass POST: HTTP 200 - 75 JSON elements.

Submission checklist

- Final PDF documentation is included.
- Request-code, retrieved-data, and JSON-response evidence is included.
- API purpose, endpoints, methods, features, and response handling are documented.
- GitHub repository link is included; no source ZIP is included by request.

GitHub repository: <https://github.com/Brenttttttt/evercare>

CONCLUSION

EverCare meets the Week 5 expected output with a working API integration that retrieves, sends, processes, and displays dynamic hospital data while handling network and response problems safely.